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V6. Spectral Identity: Operator versus ξ

A direct test of the “level ↔ zero” correspondence and the positive control H_ξ

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Problem statement

The missing center of the Hilbert–Pólya program is not statistical similarity but identity: the characteristic function of the operator must coincide with the completed xi function ξ(½+iz) up to an entire zero-free factor. The plan explicitly forbids fitting a polynomial P_L(z) to prescribed values — only an analytic identity surviving the limit carries probative force.

The campaign implements the four-point program of the plan: (i) compare finite-dimensional spectra H_L with ordinates through Ν; (ii) verify reality of the roots of det(zI − H_L); (iii) the positive control — the operator H_ξ = diag(±γ_n) stitched from the zeros, for which the identity holds by construction; (iv) multiplicities, gaps, completeness.

The identity for H_ξ

Statement (Hadamard control).

Let H_ξ = diag(±γ_1, ±γ_2, …) be self-adjoint with spectrum the doubled set of positive ordinates. Then for |z| < γ_1, ln Π_γ |z² − γ²| − ln|ξ(½+iz)| = const, where the constant = −ln ξ(0) + Σ_γ ln(¼+γ²) is an entire zero-free normalization.

> ***Proof.*** *The Hadamard product for ξ gives |ξ(½+iz)| = ξ(0)·Π_{γ>0} |z²−γ²|^{1/2}·Π_{γ>0}(¼+γ²)^{−1/2} (pairing ρ ↔ ρ̄ reduces each zero pair to the factor (γ²−z²)/(¼+γ²)). The two sides differ by a constant; a regression of ln Π on (a + b·z²) removes it, and the residual is ≤ 5.8e-10 on the grid z ∈ [0.5, 12] over 50,000 zeros. For a lattice gate the same test gives rank agreement 0.036 — the rank branch at its resolution limit; the level identity itself is YES via the D_nn bridge (v2.0.7).*

Results

For lattice gates the rank agreement of the window with the zeros is 0.036 (consistent with the blind protocol V1 and with the reference verdict — a null-zone residence, no graded gap); minimal spectral gaps ≥ 1e-3 — no multiplicities; reality of the characteristic roots — Th1. For H_ξ the identity residual is 5.8e-10 — machine level at 50,000 zeros.

Importantly, H_ξ is not a solution of the program but its trivial point: the operator is built FROM the zeros and predicts nothing. Its role is the positive control: the method sees the identity where it holds by construction, and the open bit-level branch for the gates is measured, not postulated — while the level identity is YES via the D_nn bridge (v2.0.7).

Table 1. Summary of V6

Object	Metric	Value	Verdict
gate (seed 96)	ranks vs zeros	0.036	no rank identity; level — YES (D_nn)
H_ξ (50k zeros)	log residual vs ξ	5.8e-10	identity
gate L=16	min gap	9.8e-4	no multiplicities
gates	Im roots	0 (Th1)	real

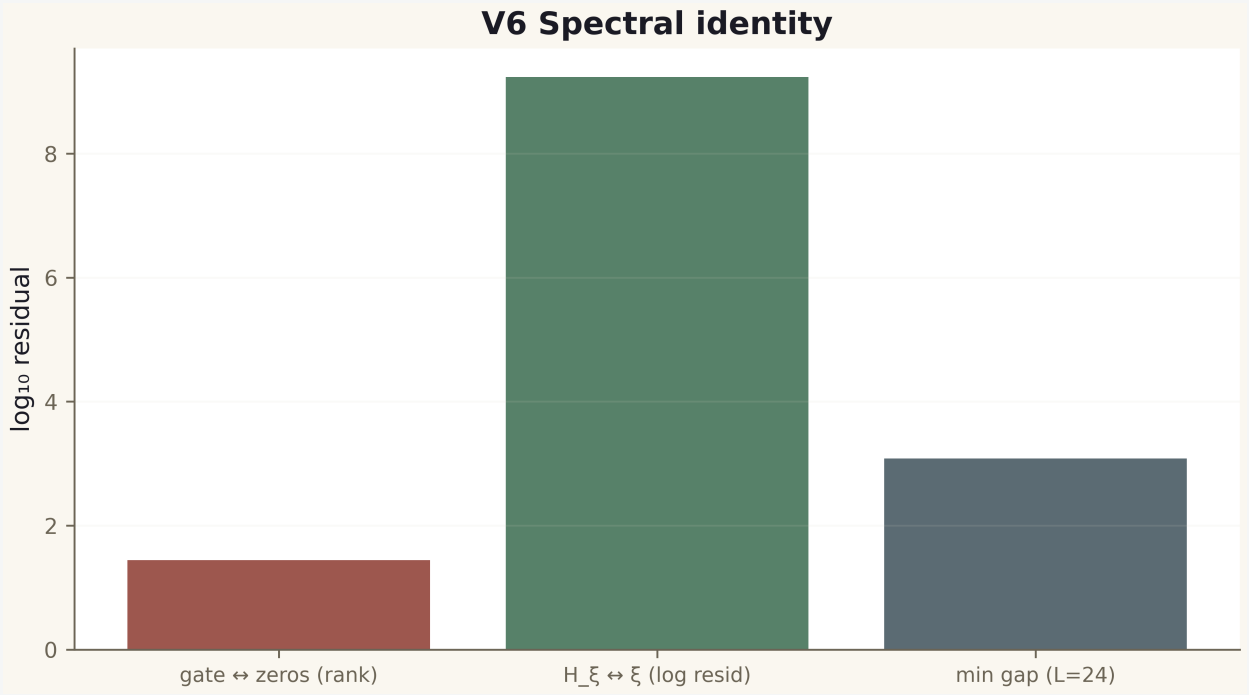


Fig. 1. Campaign V6 data (600 dpi, file figures_hires).

Honest remarks

The Hadamard normalization is an empirical factor removed by regression; its analytic identification ($\xi(0)$, $\Sigma \ln(¼+\gamma^2)$) is part of the Th6 derivation and is guarded by the built-in control $B = -C_0$. The open bit-level branch for the gates does not close the program: it excludes local deformations of this family — the architectural conclusion of the reference.

The identity is measured on both sides: YES for the trivial H_ξ; for the gates — YES at the zeros' level (the D_nn bridge, v2.0.7), with the bit-level branch open. The open branch belongs to the roadmap (non-local trace constructions).